

```
import random

# Fitness function
def fitness(x):
    return x ** 2

# Step 1: Create initial population
population = [random.randint(0, 10) for _ in range(6)]

for generation in range(5):

    print(f"\nGeneration {generation + 1}")
    print("Population:", population)

    # Step 2: Calculate fitness
    population.sort(key=fitness, reverse=True)

    print("Best value:", population[0])
    print("Best fitness:", fitness(population[0]))

    # Step 3: Select the two best parents
    parent1 = population[0]
    parent2 = population[1]

    # Step 4: Create a new population
    new_population = [parent1, parent2]
```

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while len(new_population) < 6:

    # Simple crossover: average of parents
    child = (parent1 + parent2) // 2

    # Step 5: Mutation
    mutation = random.choice([-1, 0, 1])
    child = child + mutation

    # Keep the value between 0 and 10
    child = max(0, min(10, child))

    new_population.append(child)

population = new_population
```

OUT PUT:

Generation 1

Population: [9, 6, 10, 5, 2, 3]

Best value: 10

Best fitness: 100

Generation 2

Population: [10, 9, 6, 5, 3, 2]

Best value: 10

Best fitness: 100

Generation 3

Population: [10, 9, 6, 5, 3, 2]

Best value: 10

Best fitness: 100

Generation 4

Population: [10, 9, 6, 5, 3, 2]

Best value: 10

Best fitness: 100

Generation 5

Population: [10, 9, 6, 5, 3, 2]

Best value: 10

Best fitness: 100
